

Setting up OLS

Consider a dependent variable y and vector of k independent variables $X = [x_1 \ x_2 \ \dots \ x_k]$. We observe n pairs of $(y_i, X_i)_{i=1}^n$ and hypothesize a linear relationship such that:

$$\begin{array}{rcllcll} y_1 & = & \beta_0 & + & \beta_1 x_{12} & + & \beta_2 x_{13} + \cdots + \beta_k x_{1k} + e_1 \\ y_2 & = & \beta_0 & + & \beta_1 x_{22} & + & \beta_2 x_{23} + \cdots + \beta_k x_{2k} + e_2 \\ \vdots & & \vdots & & \vdots & & \ddots & \vdots & \vdots \\ y_n & = & \beta_0 & + & \beta_1 x_{n2} & + & \beta_2 x_{n3} + \cdots + \beta_k x_{nk} + e_n \end{array}$$

This system of equations is equivalently represented by $y = X\beta + e$, where

$$\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}_{n \times 1} = \begin{bmatrix} 1 & x_{12} & x_{13} & \dots & x_{1k} \\ 1 & x_{22} & x_{23} & \dots & x_{2k} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n2} & x_{n3} & \dots & x_{nk} \end{bmatrix}_{n \times k} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{bmatrix}_{k \times 1} + \begin{bmatrix} e_1 \\ e_2 \\ \vdots \\ e_n \end{bmatrix}_{n \times 1}$$

$$y_{(n \times 1)} = X_{(n \times k)(k \times 1)} \beta + e_{(n \times 1)}$$

Deriving OLS

The least squares solution for β is the one which minimizes the sum of squared errors:

$$\beta = \underset{\beta}{\operatorname{argmin}} \|y - X\beta\|^2 = \underset{\beta}{\operatorname{argmin}} \|e\|^2 = \underset{\beta}{\operatorname{argmin}} e'e$$

We can contextualize this as finding the set of coefficients which minimize the distance between the vector of outcome observations y and vector of fitted values $X\beta$. This distance is the magnitude of e , or the square root of the sum of squared errors $e'e$.

$$\text{SSE} = e'e = \begin{bmatrix} e_1 & e_2 & \dots & e_n \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \\ \vdots \\ e_n \end{bmatrix} = \sum_{i=1}^n e_i^2$$

Deriving the least squares solution:

$$\begin{aligned} e'e &= (y - X\beta)'(y - X\beta) \\ &= y'y - y'X\beta - \beta'X'y + \beta'X'X\beta \\ &= y'y - 2\beta'X'y + \beta'X'X\beta \\ \frac{\partial e'e}{\partial \beta} &= -2X'y + 2X'X\beta = 0 \\ X'X\beta &= X'y \\ \beta &= (X'X)^{-1}X'y \end{aligned}$$

Recall that our observations are draws from a population. When we use OLS on data, we are estimating β and e from the inputs $(y_i, X_i)_{i=1}^n$. We write this as $y = X\hat{\beta} + \hat{e}$, where $\hat{\beta} = (X'X)^{-1}X'y$.

From this we also get our first “assumption”, that $X'X$ must be invertible. You know this as the assumption of no perfect collinearity. If any variable in X were a linear combination of other variables, $X'X$ would be singular and there would be infinitely many least squares solutions: the addition of any vector in the null space of $X'X$ to any $\|e\|^2$ -minimizing β is itself a $\|e\|^2$ -minimizing β .

Example

Suppose we observe 4 pairs of $(y_i, x_i) = \{(0, 1), (1, 2), (0, 2), (1, 3)\}$. Then,

$$X = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 2 & 3 \end{bmatrix}' \quad y = \begin{bmatrix} 0 & 1 & 0 & 1 \end{bmatrix}'$$

$$\hat{\beta} = (X'X)^{-1}X'y = \begin{bmatrix} 4 & 8 \\ 8 & 18 \end{bmatrix}^{-1} \begin{bmatrix} 2 \\ 5 \end{bmatrix} = \begin{bmatrix} \frac{9}{4} & -1 \\ -1 & \frac{1}{2} \end{bmatrix} \begin{bmatrix} 2 \\ 5 \end{bmatrix} = \begin{bmatrix} \frac{-1}{2} \\ \frac{1}{2} \end{bmatrix}$$

([graph of normal equations](#))

Projection

The space spanned by the columns of X , which contains all possible linear combinations of the form $X\beta$, is called the column space of X , $\text{Col}(X)$. This is a subspace in the observation space ($\mathbb{R}^n$).

OLS finds the linear combination of the columns of X which is closest to y . In other words, the closest vector in $\text{Col}(X)$ to y . This is the projection of y onto $\text{Col}(X)$.¹

Below we visualize projection in the case of two independent variables and three observations.

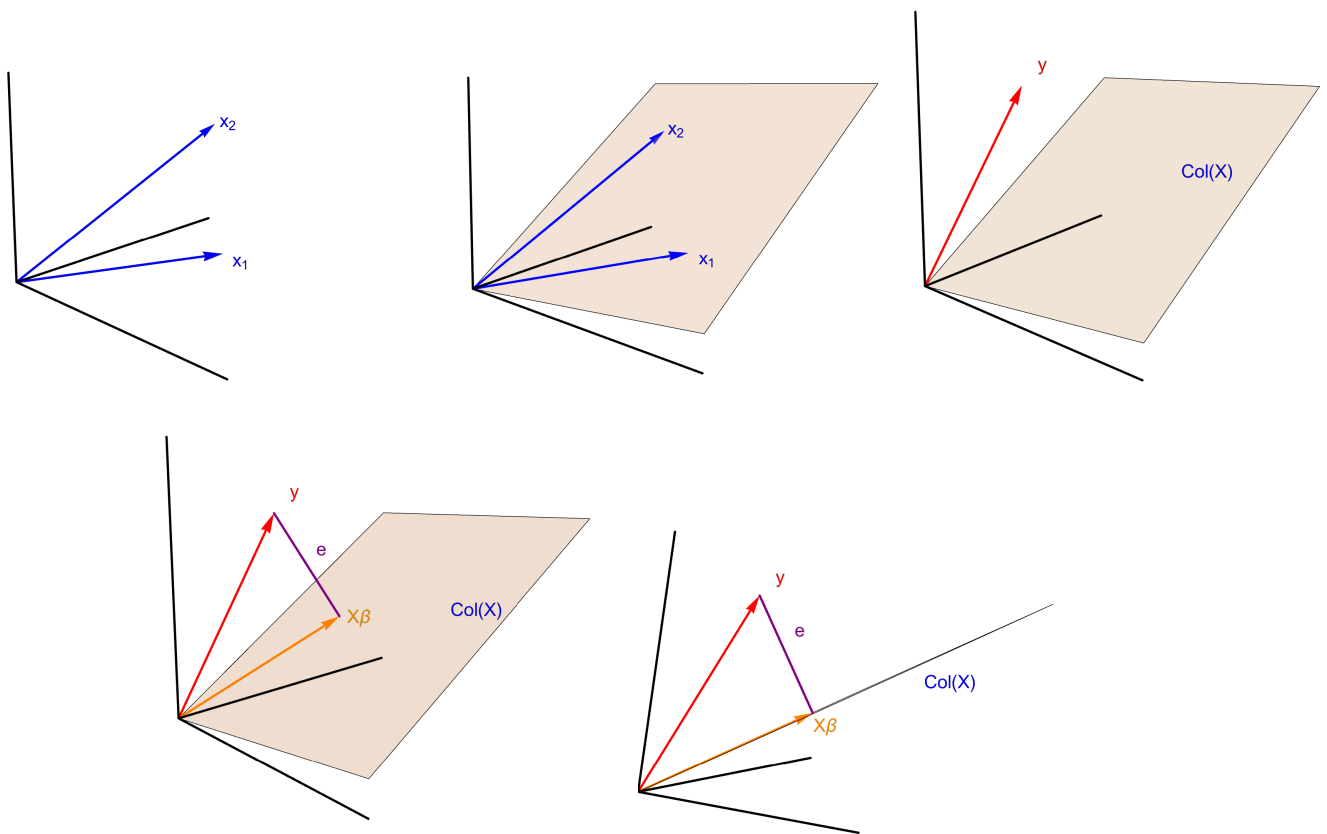


Figure 1: Projection of y onto $\text{Col}(X)$

From the fact that $X\beta$ and e are two sides of a right triangle with hypotenuse y , we get some fun properties:

1. Intuitive proof of $X\beta \perp e$ (it's a right angle!)

¹We're assuming y and X have already been de-meaned, i.e., the constant has been partialled out.

2. ANOVA decomposition follows directly from the Pythagorean theorem: $\|y\|^2 = \|X\beta\|^2 + \|e\|^2$
3. Let θ be the angle between y and $X\beta$ at the origin. The model's R^2 is the squared cosine.² By definition of cosine, $\cos^2(\theta) = (\frac{\text{adj}}{\text{hyp}})^2 = \frac{\|X\beta\|^2}{\|y\|^2} = \frac{ESS}{TSS} = R^2$
4. Intuitive proof of the convexity of the OLS solution. Any other choice of e would necessarily be longer. As you move away from e in any direction, its length increases monotonically: it's a plane.

Frisch-Waugh-Lovell theorem

Consider a partition of X into two sets of covariates such that

$$y = X_1\beta_1 + X_2\beta_2 + e$$

Theorem. *Given $y = X_1\beta_1 + X_2\beta_2 + e$. The OLS solution for β_1 and e can be obtained with the three-step process of:*

1. *Regress y on X_2 , saving residuals y^**
2. *Regress each variable in X_1 on X_2 , saving residuals X_1^**
3. *Regress y^* on X_1^* , obtaining β_1 and e*

Proof. Consider the residual-maker matrix $M_{X_2} = I - X_2(X_2'X_2)^{-1}X_2'$. Pre-multiply both sides of the regression equation by M_{X_2} :

$$\begin{aligned} M_{X_2}y &= M_{X_2}(X_1\beta_1 + X_2\beta_2 + e) \\ M_{X_2}y &= M_{X_2}X_1\beta_1 + M_{X_2}X_2\beta_2 + M_{X_2}e \\ y^* &= X_1^*\beta_1 + e \end{aligned}$$

It follows from the OLS derivation that β_1 and e can be obtained by the regression of y^* on X_1^* . □

²An aside: the cosine of the angle between two variables' observation vectors is their correlation. Let a and b be mean-zero vectors of observations with interior angle α . Then, $\text{corr}(a, b) = \frac{\text{cov}(a, b)}{\sigma_a\sigma_b} = \frac{a'b}{\|a\|\|b\|} = \cos(\alpha)$.

Bias

Theorem. Given $y = X\beta + e$, $X'X$ invertible, and $\mathbb{E}[e|X] = 0$, the OLS estimator $\hat{\beta} = (X'X)^{-1}X'y$ is conditionally unbiased for β .

Proof.

$$\begin{aligned}
 \mathbb{E}[\hat{\beta}|X] &= \mathbb{E}[(X'X)^{-1}X'y|X] \\
 &= (X'X)^{-1}X'\mathbb{E}[y|X] \\
 &= (X'X)^{-1}X'\mathbb{E}[X\beta + e|X] \\
 &= (X'X)^{-1}X'X\beta + (X'X)^{-1}X'\mathbb{E}[e|X] \\
 &= \beta
 \end{aligned}$$

□

Consider the two regressions $y = X_1\beta_1 + X_2\beta_2 + e$ and $y = X_1\tilde{\beta}_1 + \tilde{e}$. We seek to express $\tilde{\beta}_1$ in terms of $\hat{\beta}_1$ plus some bias term.

$$\begin{aligned}
 \tilde{\beta}_1 &= (X_1'X_1)^{-1}X_1'y \\
 &= (X_1'X_1)^{-1}X_1'(X_1\hat{\beta}_1 + X_2\hat{\beta}_2 + e) \\
 &= (X_1'X_1)^{-1}X_1'X_1\hat{\beta}_1 + (X_1'X_1)^{-1}X_1'X_2\hat{\beta}_2 + (X_1'X_1)^{-1}X_1'e \\
 &= \hat{\beta}_1 + (X_1'X_1)^{-1}X_1'X_2\hat{\beta}_2
 \end{aligned}$$

Note that $(X_1'X_1)^{-1}X_1'X_2$ is a $k \times p$ matrix of regression coefficients, where each column is the set of coefficients from the regression of the p th variable of X_2 on the k variables of X_1 .

Looking at the i th individual coefficient in $\tilde{\beta}_1$, which we'll call $\tilde{b}_{1i}$, its bias can be expressed as $\tilde{b}_{1i} = b_{1i} + \sum_{j=1}^p \alpha_{ij}b_{2j}$, where α_{ij} is the coefficient on x_{1i} in the regression of x_{2j} on X_1 .

Each variable x_{2p} in X_2 contributes a term in the bias of each coefficient $\tilde{b}_{1i}$ in $\tilde{\beta}_1$. Each term is the product of two regression coefficients: the coefficient on x_{1i} in the regression of x_{2p} on X_1 , and the coefficient on x_{2p} in the full regression of y on X_1 and X_2 . The main takeaways from this are:

1. Adding an additional variable to X_1 can affect the bias associated with each omitted variable by influencing the α_{ij}

2. Adding an additional variable to X_2 can affect the bias associated with each omitted variable by influencing the b_{2j}
3. Including more variables in X_1 does not necessarily decrease the magnitude of bias

Deriving variance (under homoskedasticity)

Let's look at the outer product ee' . This is an $n \times n$ matrix where the diagonal is the square of the i th error, and the off-diagonals are the product of each e_i with each e_j . We assume that $\mathbb{E}[ee'|X] = \mathbb{E}[ee'] = \sigma^2 I$, or:

1. X tells us nothing about the error variance ($\mathbb{E}[ee'|X] = \mathbb{E}[ee']$)
2. All errors have the same variance σ^2 (the elements on the diagonal are all σ^2)
3. Errors are uncorrelated between observations: the expectation of the product of any two errors e_i and e_j , $i \neq j$, is 0 (the elements on the off-diagonal of $\mathbb{E}[ee']$ are all 0s)

Theorem. *Given that $\mathbb{E}[\hat{\beta}|X] = \beta$ and $\mathbb{E}[ee'|X] = \mathbb{E}[ee'] = \sigma^2 I$, the OLS estimator $\hat{\beta} = (X'X)^{-1}X'y$ has conditional variance $\sigma^2(X'X)^{-1}$.*

Proof.

$$\begin{aligned}
 \text{var}[\hat{\beta}|X] &= \mathbb{E}[(\hat{\beta} - \mathbb{E}[\hat{\beta}|X])(\hat{\beta} - \mathbb{E}[\hat{\beta}|X])'|X] \\
 &= \mathbb{E}[(\hat{\beta} - \beta)(\hat{\beta} - \beta)'|X] \\
 &= \mathbb{E}[(X'X)^{-1}X'ee'X(X'X)^{-1}|X] \\
 &= (X'X)^{-1}X'\mathbb{E}[ee'|X]X(X'X)^{-1} \\
 &= (X'X)^{-1}X'\sigma^2 IX(X'X)^{-1} \\
 &= \sigma^2(X'X)^{-1}X'X(X'X)^{-1} \\
 &= \sigma^2(X'X)^{-1}
 \end{aligned}$$

□

Ridge regression

Consider a scientist interested in predicting an outcome variable with so many predictor variables that they have more variables than observations ($k > n$). In this case, $X'X$ is not invertible and there are infinitely many least squares solutions. Picking coefficients randomly isn't going to give this scientist a model with strong out-of-sample performance.

Ridge regression is an option in cases like these: it imposes a penalty, or shrinkage, on the magnitude of the coefficient vector β . This ensures that there is a solution, and the strength of this penalty can be tuned to optimize for out-of-sample performance. It doesn't have the same kind of interpretation as OLS (ridge is biased), but it does help with prediction, assigning less-useful variables low-magnitude coefficients and more-useful variables higher-magnitude coefficients.

Compare the OLS and ridge regression loss functions:

$$\begin{aligned}\beta_{\text{OLS}} &= \underset{\beta}{\operatorname{argmin}} \|y - X\beta\|^2 \\ \beta_{\text{ridge}} &= \underset{\beta}{\operatorname{argmin}} \|y - X\beta\|^2 + \lambda\|\beta\|^2\end{aligned}$$

There are many algorithms to select λ , but typically the goal is to maximize out-of-sample performance. It's important to note that the ridge estimator is sensitive to changes in scale, so many programs first standardize each variable before estimating β_{ridge} .

Deriving the ridge solution from its loss function $L(\beta_{\text{ridge}})$:

$$\begin{aligned}L(\beta_{\text{ridge}}) &= \|y - X\beta\|^2 + \lambda\|\beta\|^2 \\ &= (y - X\beta)'(y - X\beta) + \lambda\beta'\beta \\ &= y'y - 2\beta'X'y + \beta'X'X\beta + \lambda\beta'\beta \\ \frac{\partial L(\beta_{\text{ridge}})}{\partial \beta} &= -2X'y + 2X'X\beta + 2\lambda\beta = 0 \\ X'X\beta + \lambda\beta &= X'y \\ (X'X + \lambda I)\beta &= X'y \\ \beta &= (X'X + \lambda I)^{-1}X'y\end{aligned}$$

This is very similar to the OLS derivation, but with the shrinkage/penalty term.

A fun fact: many algorithms for computing OLS estimates can be turned into ridge regression algorithms by appending $\sqrt{\lambda}I$ to the bottom of X (and, if needed, the corresponding amount of 0s to y).

[Ridge normal equations in Desmos.](#)